

Lecture 10: Image Segmentation

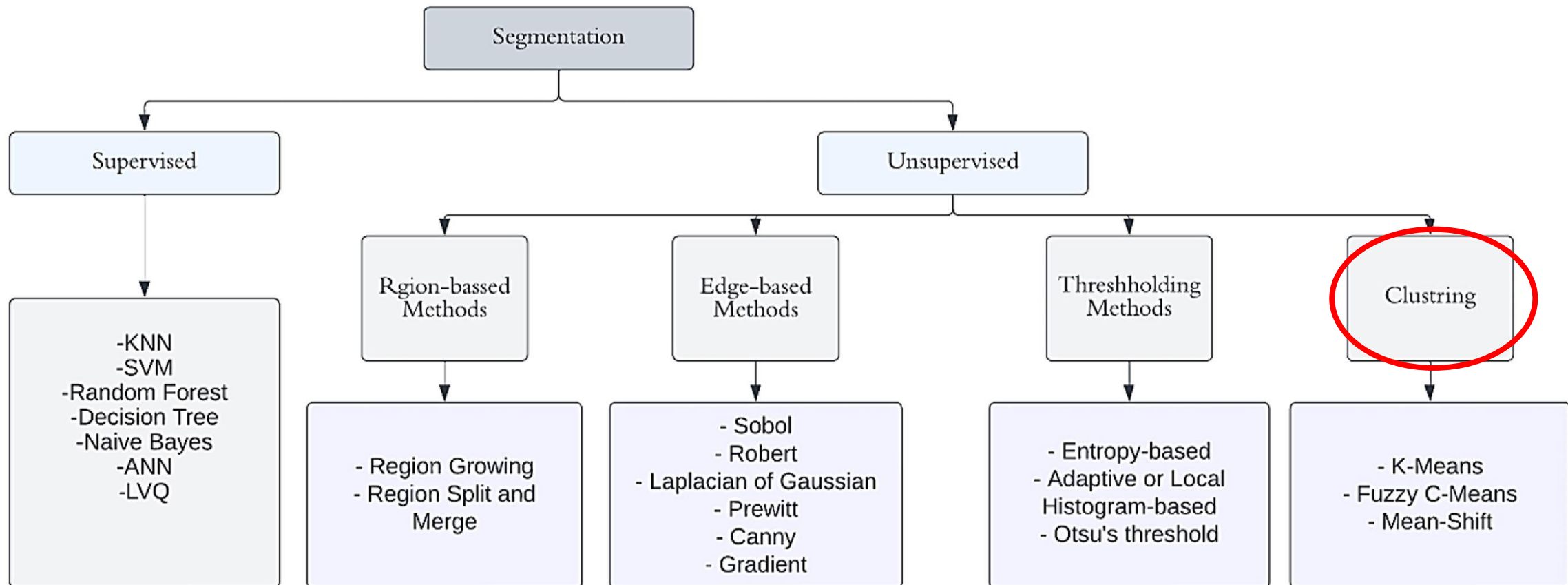
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Image segmentation algorithms

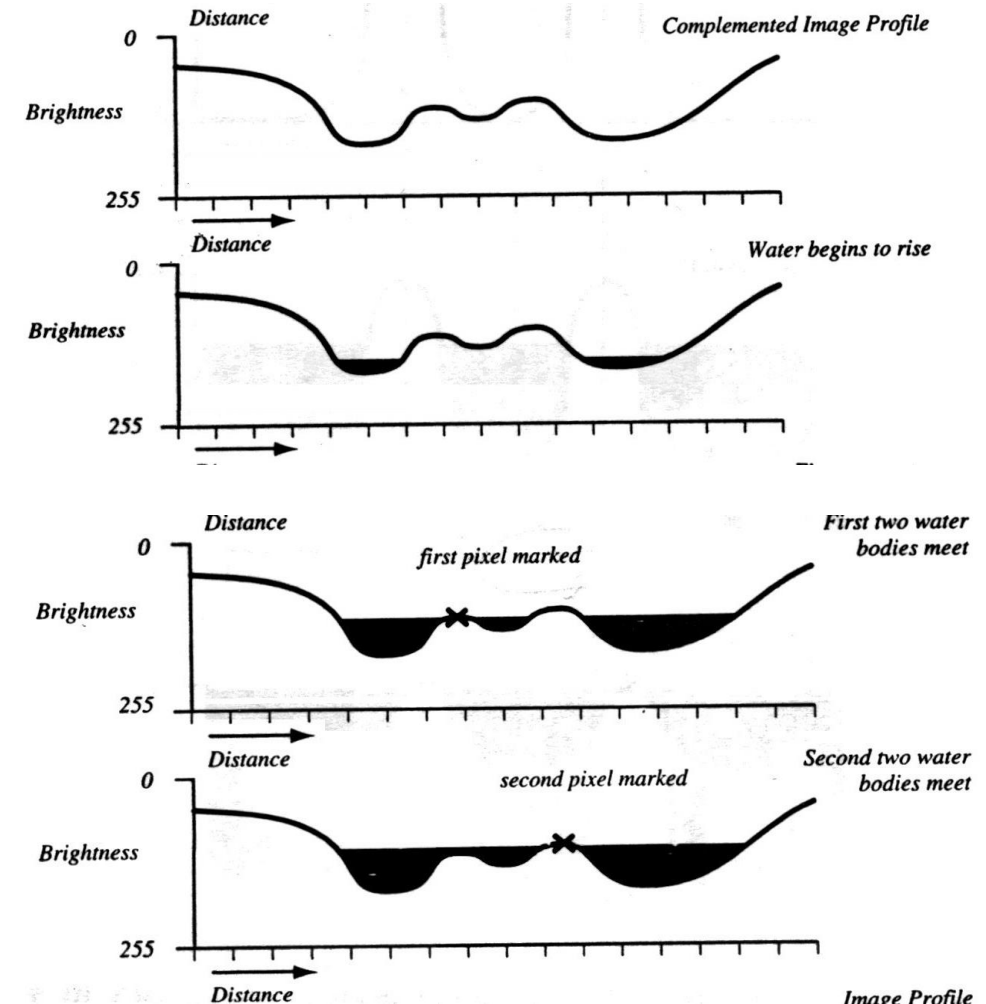
- Edge– based methods
 - Point, line, edge detection filters
 - High-level edge detection algorithms
- Region-based methods
 - Thresholding
 - Region growing
- Watershed
- Clustering
 - k-mean
 - SLIC
- Application

Overview



Watershed Segmentation

- To find watershed lines.
 - Suppose that a hole is punched in each regional minimum
 - The entire topography is flooded from below by letting water rise through the holes at a uniform rate.
 - When rising water in distinct catchment basins is about the merge, a dam is built to prevent merging.
 - These dam boundaries correspond to the watershed lines.



Watershed Segmentation Algorithm

- Start with all pixels with the lowest possible value.
 - These form the basis for initial watersheds
- For each intensity level k :
 - For each group of pixels of intensity k
 - If adjacent to exactly one existing region, add these pixels to that region
 - Else if adjacent to more than one existing regions, mark as boundary
 - Else start a new region

Watershed example

8	7	5	4	3
9	6	2	1	2
4	2	7	3	4
3	0	1	8	2
6	1	3	6	1

image

8	7	5	4	3
9	6	2	1	2
4	2	7	3	4
3	0	1	8	2
6	1	3	6	1

image

Step 1: Compute pixel list

- 0: (1,3)
- 1:
- 2:
- 3:
- 4:
- 5:
- 6:
- 7:
- 8:
- 9:

pixel list

Watershed example

8	7	5	4	3
9	6	2	1	2
4	2	7	3	4
3	0	1	8	2
6	1	3	6	1

image

Step 1: Compute pixel list

- 0: (1,3)
- 1: (3,1), (2,3), (1,4), (4,4)
- 2: (2,1), (4,1), (1,2), (4,3)
- 3: (4,0), (3,2), (0,3), (2,4)
- 4: (3,0), (0,2), (4,2)
- 5: (2,0)
- 6: (1,1), (0,4), (3,4)
- 7: (1,0), (2,2)
- 8: (0,0), (3,3)
- 9: (0,1)

pixel list

Watershed example

8	7	5	4	3
9	6	2	1	2
4	2	7	3	4
3	0	1	8	2
6	1	3	6	1

image

u	u	u	u	u
u	u	u	u	u
u	u	u	u	u
u	u	u	u	u
u	u	u	u	u

labels

- 0: (1,3)
- 1: (3,1), (2,3), (1,4), (4,4)
- 2: (2,1), (4,1), (1,2), (4,3)
- 3: (4,0), (3,2), (0,3), (2,4)
- 4: (3,0), (0,2), (4,2)
- 5: (2,0)
- 6: (1,1), (0,4), (3,4)
- 7: (1,0), (2,2)
- 8: (0,0), (3,3)
- 9: (0,1)

pixel list

Step 2:
Set all pixels
to “unlabeled”

Watershed example

8	7	5	4	3
9	6	2	1	2
4	2	7	3	4
3	0	1	8	2
6	1	3	6	1

image

u	u	u	u	u
u	u	u	u	u
u	u	u	u	u
u	0	u	u	u
u	u	u	u	u

labels

0: (1,3)

1: (3,1), (2,3), (1,4), (4,4)

2: (2,1), (4,1), (1,2), (4,3)

3: (4,0), (3,2), (0,3), (2,4)

4: (3,0), (0,2), (4,2)

5: (2,0)

6: (1,1), (0,4), (3,4)

7: (1,0), (2,2)

8: (0,0), (3,3)

9: (0,1)

pixel list

Step : k=0

Create new catchment
basins

Watershed example

8	7	5	4	3
9	6	2	1	2
4	2	7	3	4
3	0	1	8	2
6	1	3	6	1

image

u	u	u	u	u
u	u	u	u	u
u	u	u	u	u
u	0	0	u	u
u	0	u	u	u

labels

(shaded pixels are on frontier)

- 0: (1,3)
- 1: (3,1), (2,3), (1,4), (4,4)
- 2: (2,1), (4,1), (1,2), (4,3)
- 3: (4,0), (3,2), (0,3), (2,4)
- 4: (3,0), (0,2), (4,2)
- 5: (2,0)
- 6: (1,1), (0,4), (3,4)
- 7: (1,0), (2,2)
- 8: (0,0), (3,3)
- 9: (0,1)

pixel list

Step : k=1

Grow catchment basins

Watershed example

8	7	5	4	3
9	6	2	1	2
4	2	7	3	4
3	0	1	8	2
6	1	3	6	1

image

u	u	u	u	u
u	u	u	u	u
u	u	u	u	u
u	0	0	u	u
u	0	u	u	u

labels

(shaded pixels are on frontier)

- 0: (1,3)
- 1: (3,1), (2,3), (1,4), (4,4)
- 2: (2,1), (4,1), (1,2), (4,3)
- 3: (4,0), (3,2), (0,3), (2,4)
- 4: (3,0), (0,2), (4,2)
- 5: (2,0)
- 6: (1,1), (0,4), (3,4)
- 7: (1,0), (2,2)
- 8: (0,0), (3,3)
- 9: (0,1)

pixel list

Step 7: k=1
Expand frontier
(nowhere to go)

Watershed example

8	7	5	4	3
9	6	2	1	2
4	2	7	3	4
3	0	1	8	2
6	1	3	6	1

image

u	u	u	u	u
u	u	u	1	u
u	u	u	u	u
u	0	0	u	u
u	0	u	u	2

labels

- 0: (1,3)
- 1: (3,1), (2,3), (1,4), (4,4)
- 2: (2,1), (4,1), (1,2), (4,3)
- 3: (4,0), (3,2), (0,3), (2,4)
- 4: (3,0), (0,2), (4,2)
- 5: (2,0)
- 6: (1,1), (0,4), (3,4)
- 7: (1,0), (2,2)
- 8: (0,0), (3,3)
- 9: (0,1)

pixel list

Step 8: k=1
Create new basins

Watershed example

8	7	5	4	3
9	6	2	1	2
4	2	7	3	4
3	0	1	8	2
6	1	3	6	1

image

u	u	u	u	u
u	u	1	1	1
u	0	u	u	u
u	0	0	u	2
u	0	u	u	2

labels

(shaded pixels are on frontier)

- 0: (1,3)
- 1: (3,1), (2,3), (1,4), (4,4)
- 2: (2,1), (4,1), (1,2), (4,3)
- 3: (4,0), (3,2), (0,3), (2,4)
- 4: (3,0), (0,2), (4,2)
- 5: (2,0)
- 6: (1,1), (0,4), (3,4)
- 7: (1,0), (2,2)
- 8: (0,0), (3,3)
- 9: (0,1)

pixel list

Step 9: k=2
Grow basins

Watershed example

(shaded pixels are on frontier)

8	7	5	4	3
9	6	2	1	2
4	2	7	3	4
3	0	1	8	2
6	1	3	6	1

image

u	u	u	u	u
u	u	1	1	1
u	0	u	u	u
u	0	0	u	2
u	0	u	u	2

labels

- 0: (1,3)
- 1: (3,1), (2,3), (1,4), (4,4)
- 2: (2,1), (4,1), (1,2), (4,3)
- 3: (4,0), (3,2), (0,3), (2,4)
- 4: (3,0), (0,2), (4,2)
- 5: (2,0)
- 6: (1,1), (0,4), (3,4)
- 7: (1,0), (2,2)
- 8: (0,0), (3,3)
- 9: (0,1)

pixel list

Step 10: k=2
Expand frontier
(nowhere to go)

Watershed example

8	7	5	4	3
9	6	2	1	2
4	2	7	3	4
3	0	1	8	2
6	1	3	6	1

image

u	u	u	u	u
u	u	1	1	1
u	0	u	u	u
u	0	0	u	2
u	0	u	u	2

labels

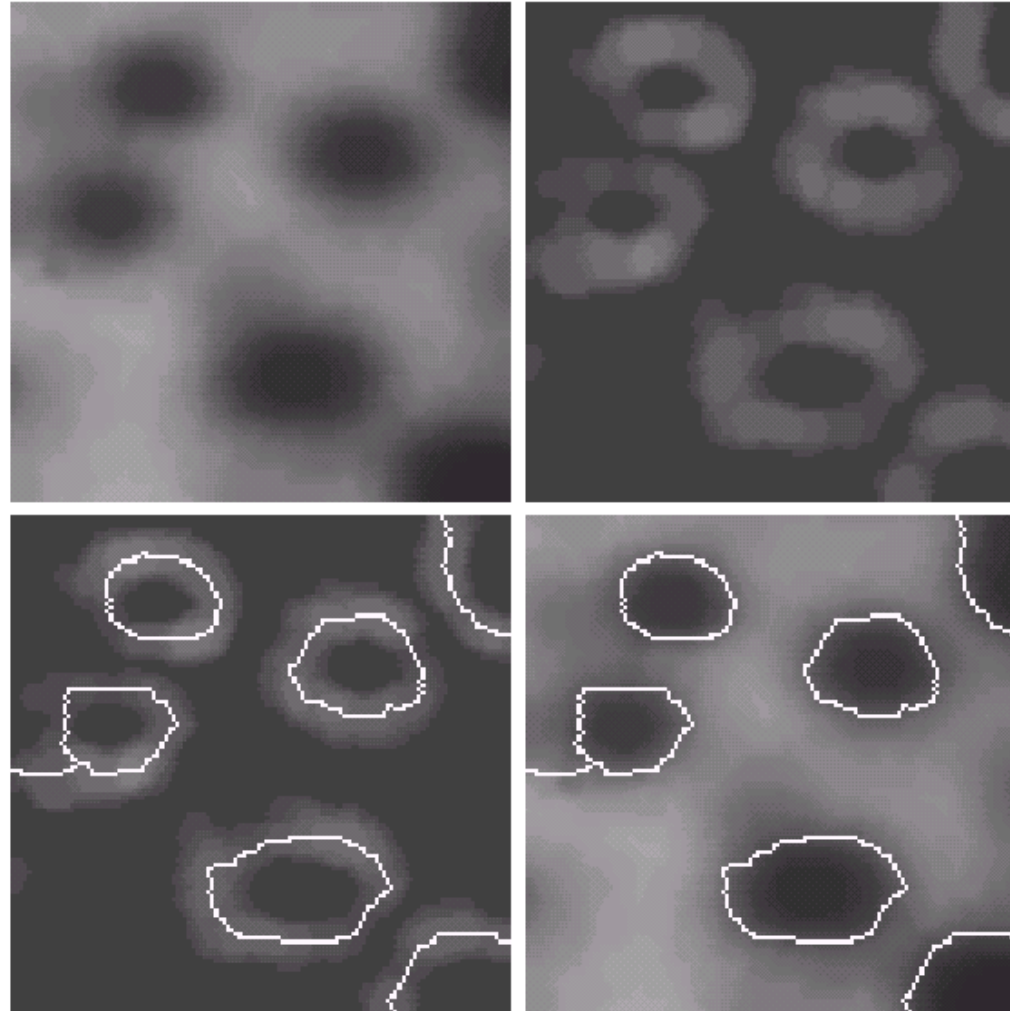
- 0: (1,3)
- 1: (3,1), (2,3), (1,4), (4,4)
- 2: (2,1), (4,1), (1,2), (4,3)
- 3: (4,0), (3,2), (0,3), (2,4)
- 4: (3,0), (0,2), (4,2)
- 5: (2,0)
- 6: (1,1), (0,4), (3,4)
- 7: (1,0), (2,2)
- 8: (0,0), (3,3)
- 9: (0,1)

pixel list

Step 11: k=2
Create new basins
(none to create)

Watershed results

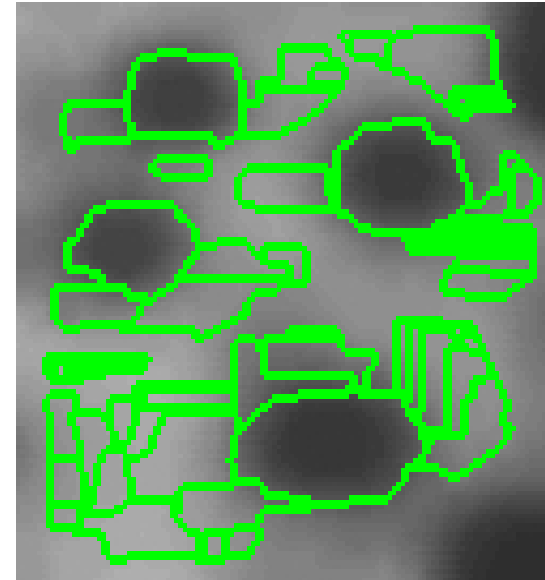
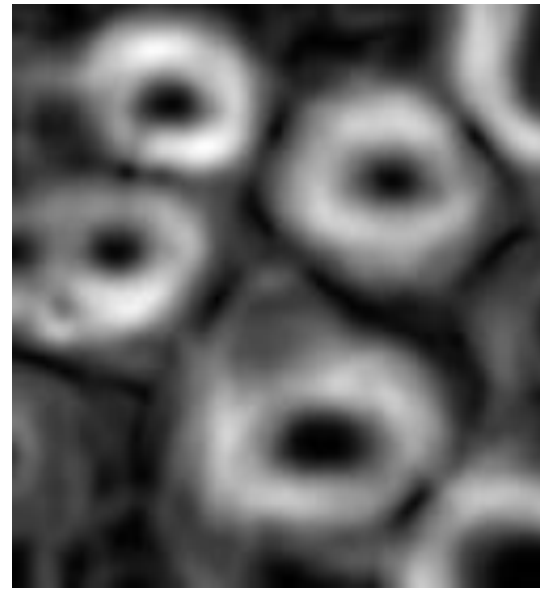
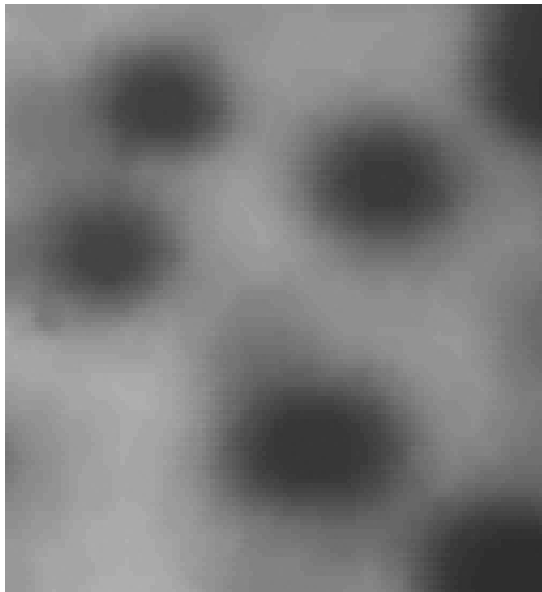
Watershed algorithm might be used on the **gradient image** instead of the original image.



Gradient
image

Noise effect on watershed

Due to noise and other local irregularities of the gradient, **oversegmentation** might occur.

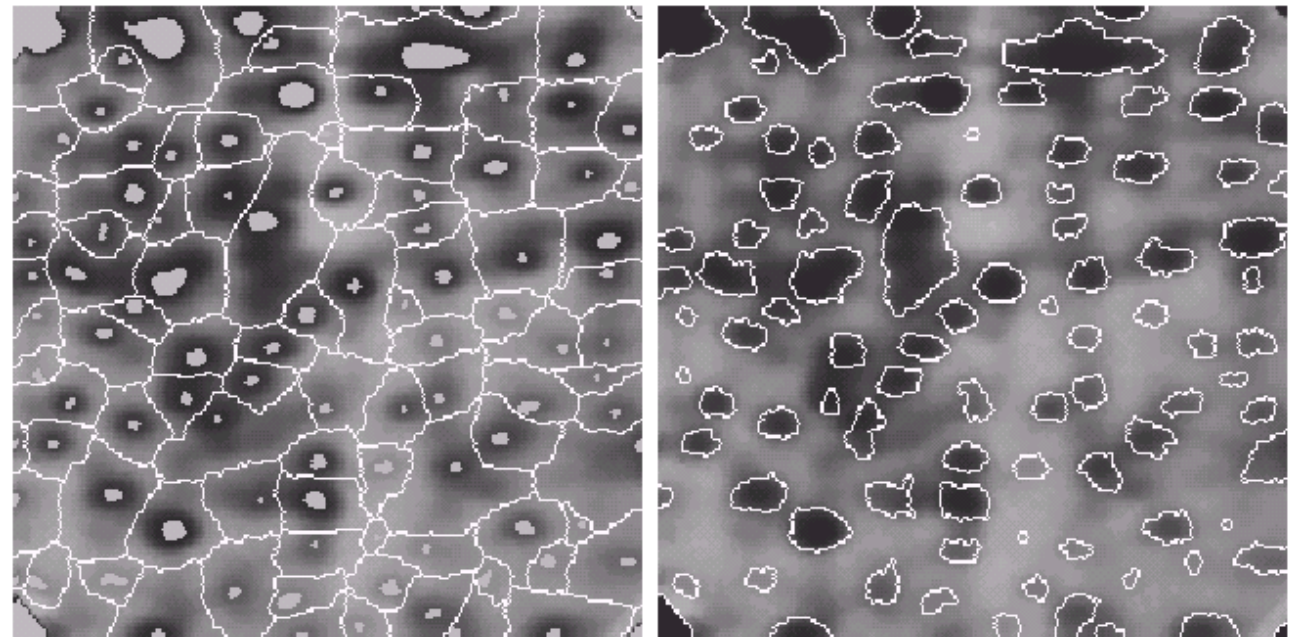


The Use of Markers

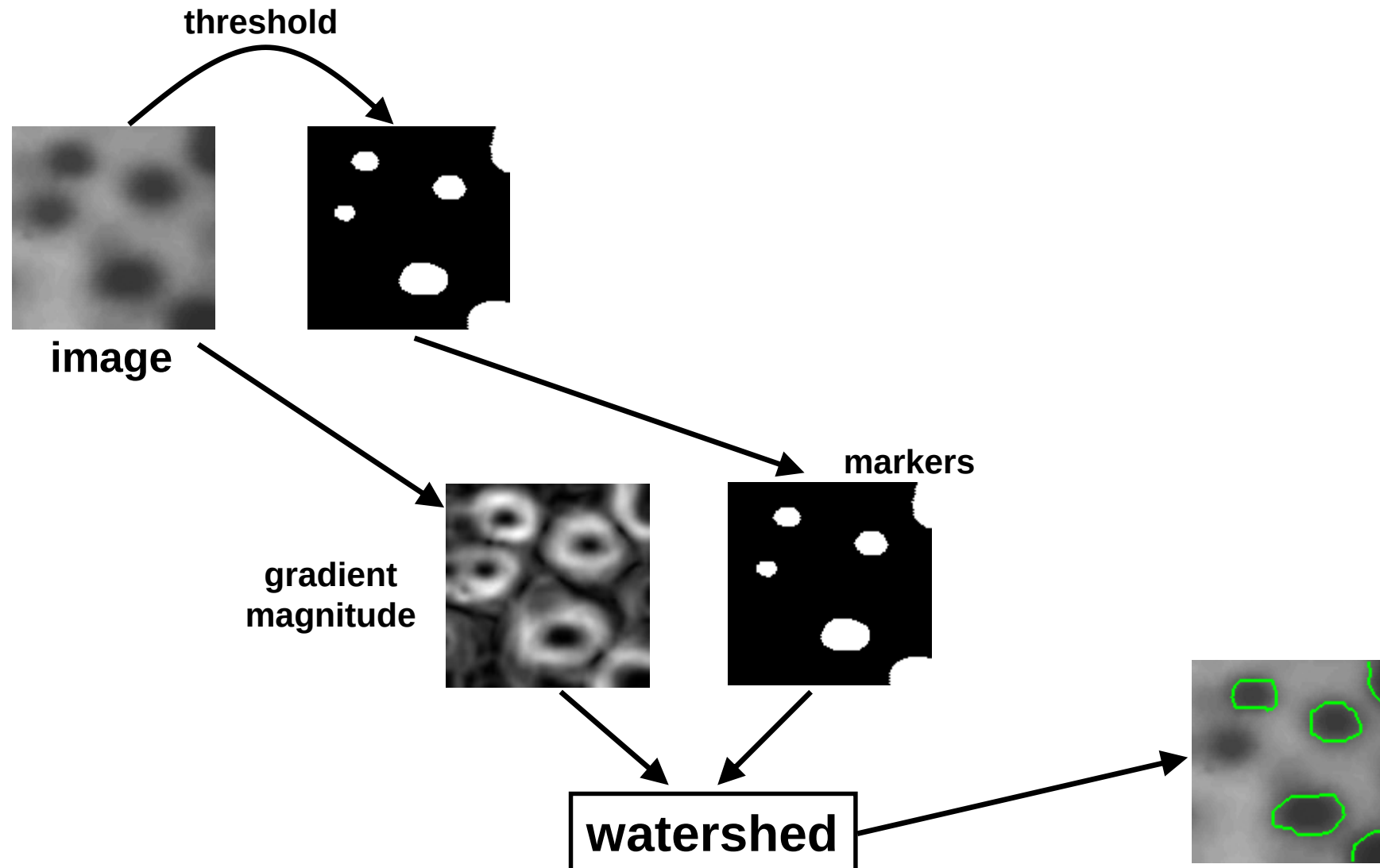
A solution is to limit the number of regional minima.
Use markers to specify the only **allowed regional minima**

Internal marker: define regions
(1) surrounded **higher** “altitude”
(2) form a **connected** component
(3) **same** intensity value.

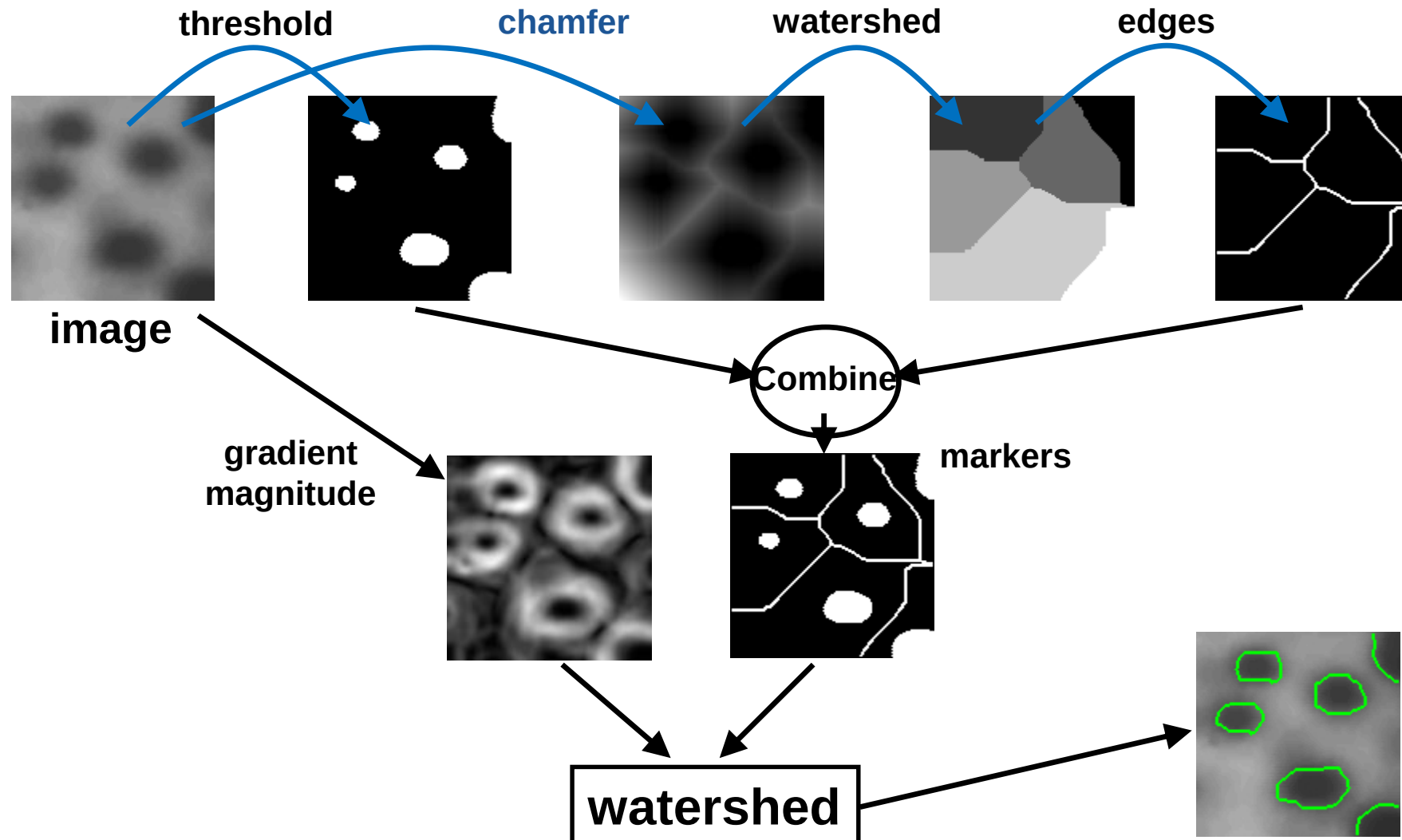
External markers: lines
a single object and background
segmentation



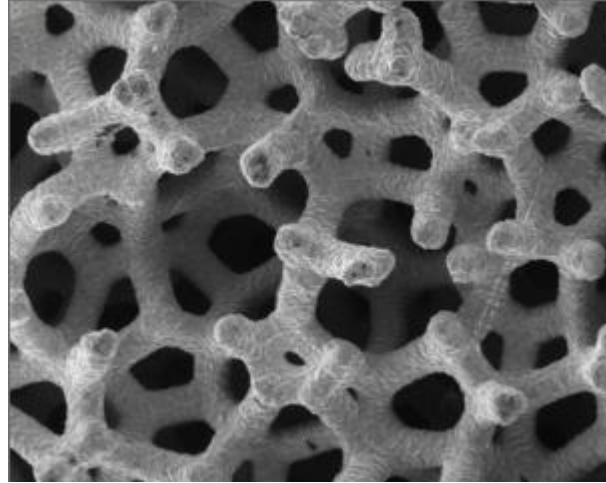
The Use of Markers



The Use of Markers



Another example



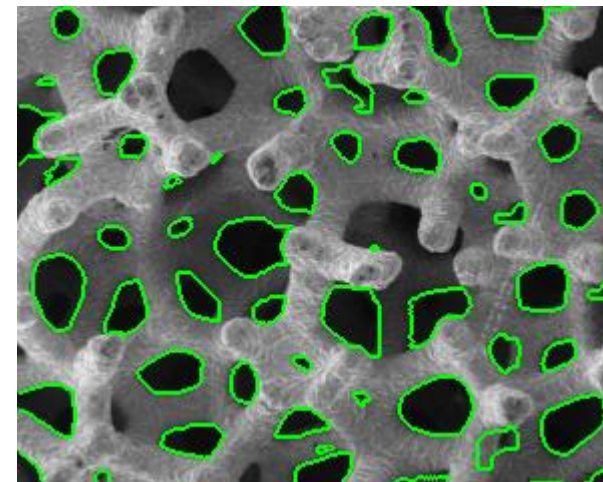
image



markers



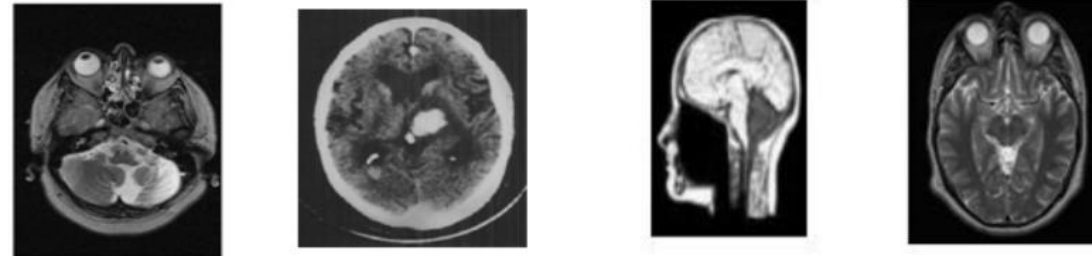
gradient magnitude



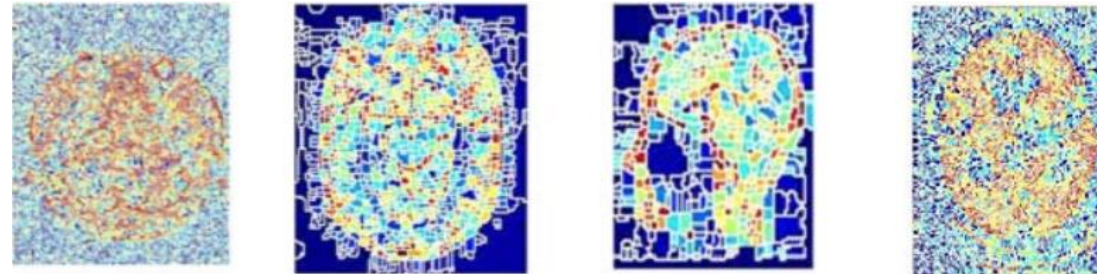
watershed

Examples in Medical Images

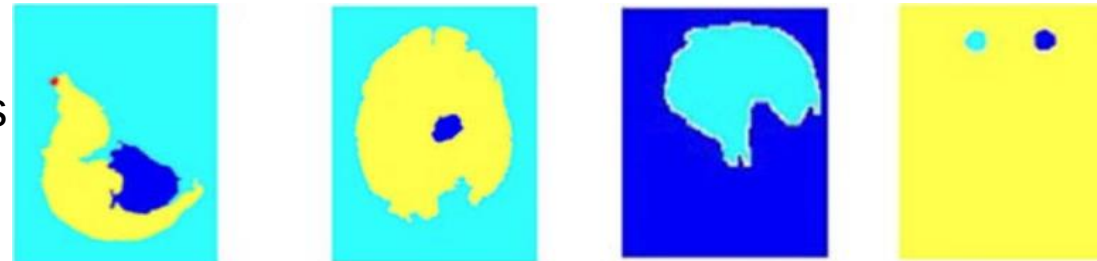
Input Image



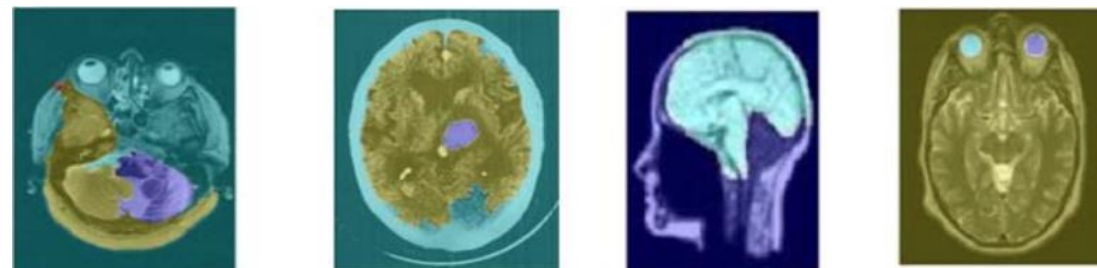
Oversegmented image
(Apply watershed without
marking objects)



Apply watershed after marks
the ROI and background



Final colored image



Clustering

- There are K clusters $C_1, \dots, C_k$ with means $m_1, \dots, m_k$.
- The **least-squares error** is defined as

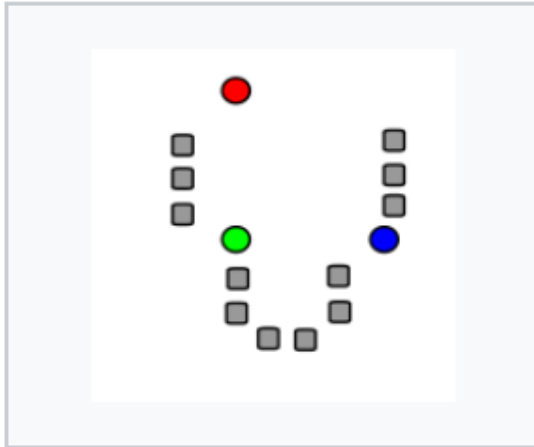
$$D = \sum_{k=1}^K \sum_{x_i \in C_k} (x_i - m_k)^2$$

- Out of all possible partitions into K clusters, choose the one that minimizes D .

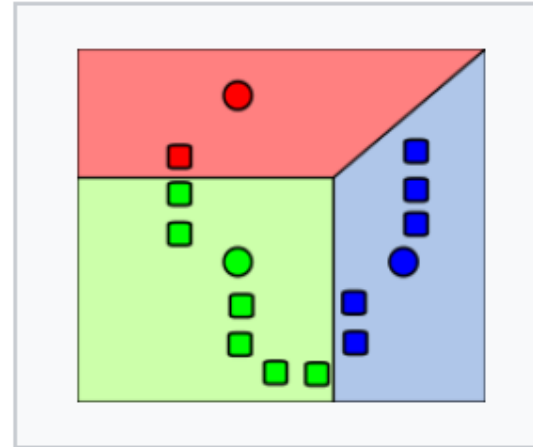
K-Means Clustering

- Form K-means clusters from a set of n-dimensional vectors
 - 1. Set ic (iteration count) to 1
 - 2. Choose randomly a set of K means $m_1(1), \dots, m_K(1)$.
 - 3. For each vector x_i compute $D(x_i, m_k(ic))$, $k = 1, \dots, K$ and assign x_i to the cluster C_j with nearest mean.
 - 4. Increment ic by 1, update the means to get $m_1(ic), \dots, m_K(ic)$.
 - 5. Repeat steps 3 and 4 until $C_k(ic) = C_k(ic + 1)$ for all k .

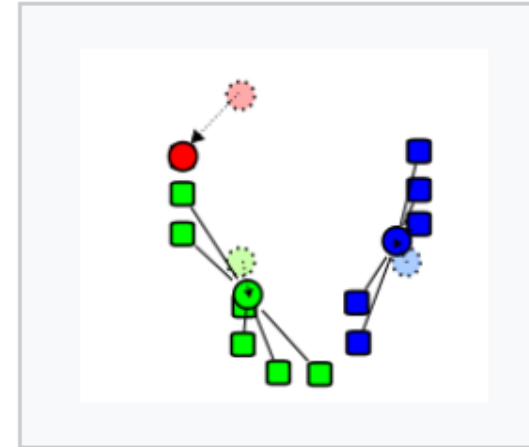
K-Means Clustering



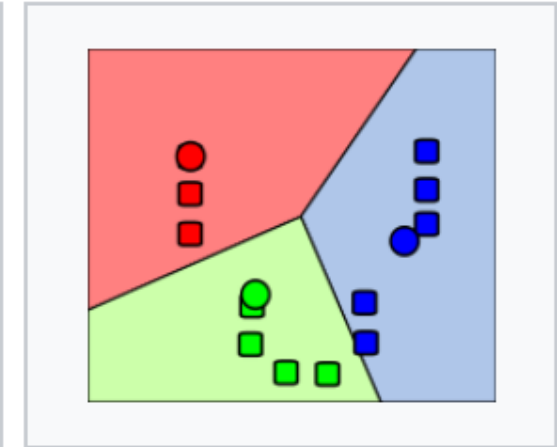
1. k initial "means" (in this case $k=3$) are randomly generated within the data domain (shown in color).



2. k clusters are created by associating every observation with the nearest mean. The

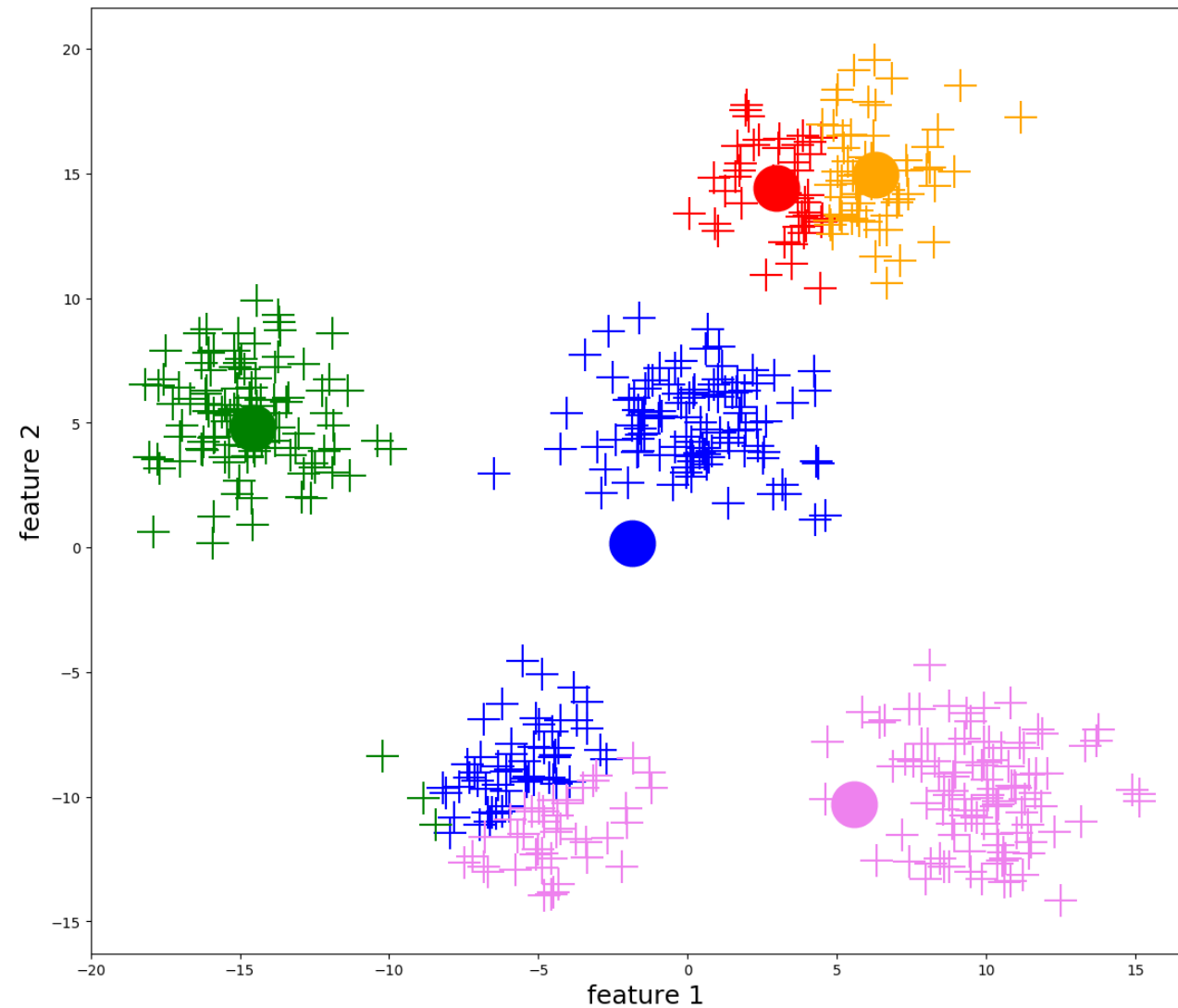


3. The **centroid** of each of the k clusters becomes the new mean.



4. Steps 2 and 3 are repeated until convergence has been reached.

K-Means Clustering



K-Means Clustering

- Algorithm

1. Initialize the algorithm:

$$m_i(1), i = 1, 2, \dots, k$$

2. Assign samples to clusters:

$$z_q \rightarrow C_i, \text{ if } (z_q - m_i)^2 < (z_q - m_j)^2, j = 1, 2, \dots, k (j \neq i); q = 1, 2, \dots, Q$$

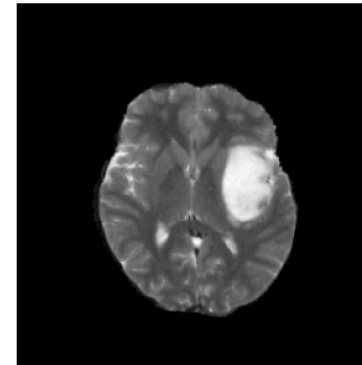
3. Update the cluster centers (means):

$$m_i = \frac{1}{|C_i|} \sum_{z \in C_i} z, i = 1, 2, \dots, k$$

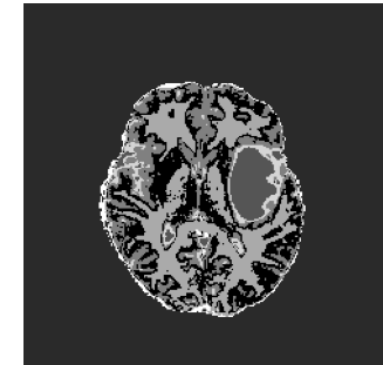
4. Test for completion:

Compute the Euclidean norms of the differences between the mean vectors in the current and previous steps. Compute the residual error, E, as the sum of the k norms. Stop if where T a specified, nonnegative threshold. Else, go back to Step 2.

Original



KMeans Segmentation k = 7



K-Means Clustering

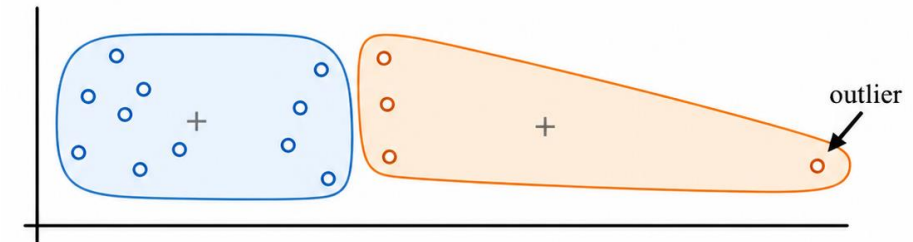
- K-means: pros and cons

Pros

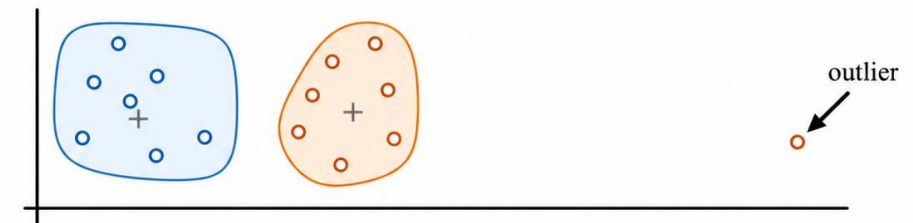
- Simple, fast to compute
- Converges to local minimum of within-cluster squared error

Cons/issues

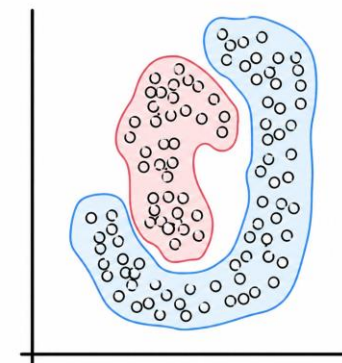
- Setting k ?
- Sensitive to initial centers
- Sensitive to outliers
- Detects spherical clusters



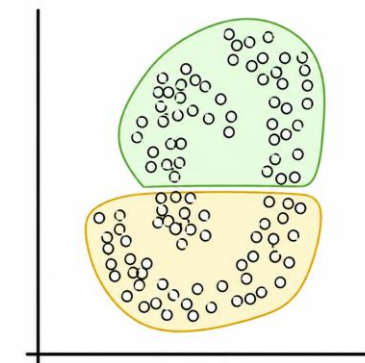
(A): Undesirable clusters



(B): Ideal clusters



(A): Two natural clusters

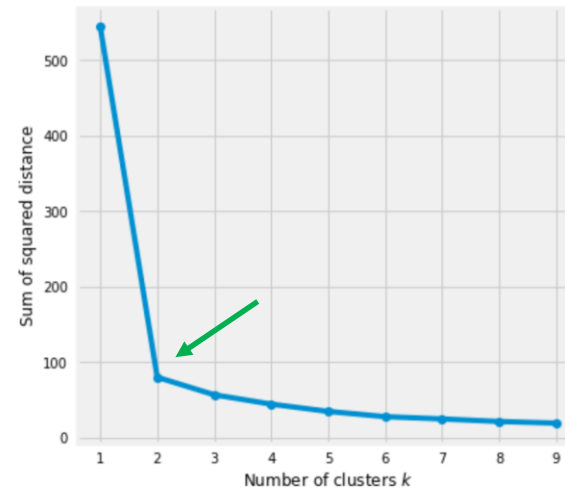


(B): k -means clusters

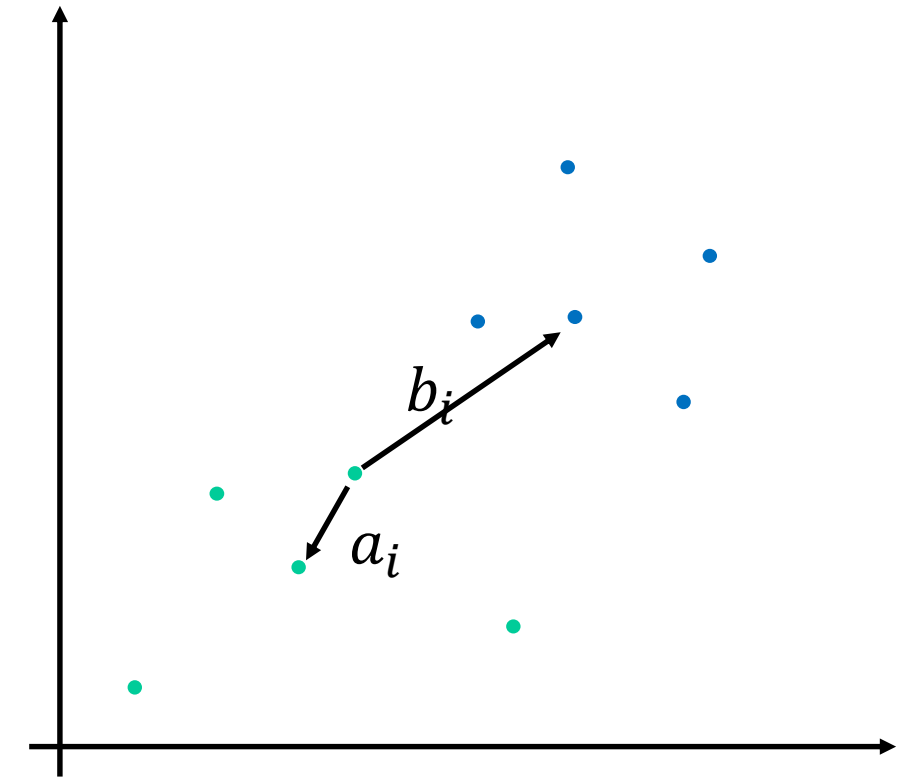
Evaluation of K-mean

- Criteria: the sum of squared distance

- L-Curve



- Silhouette Analysis
$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)}$$



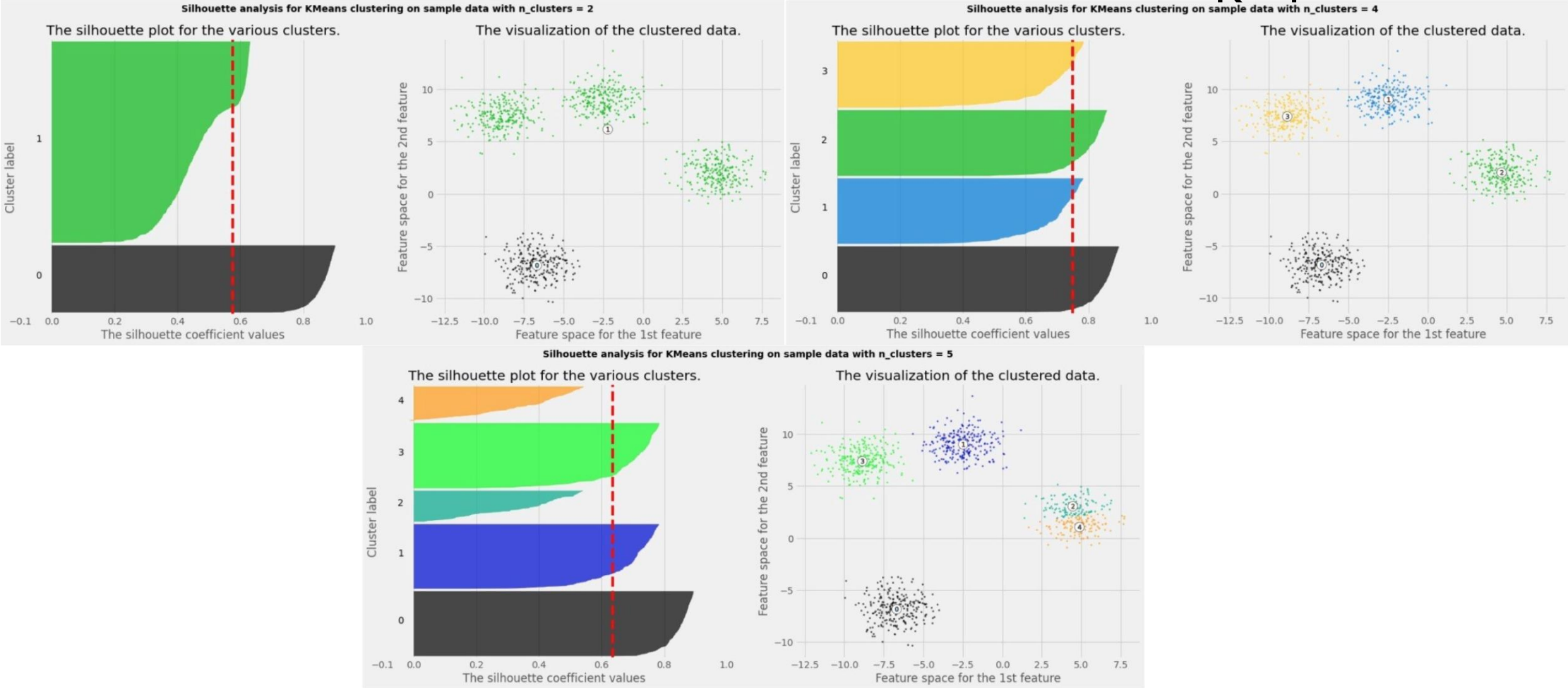
Cohesion (a_i): The average distance between a data point and all other points in the same cluster.

Separation (b_i): The average distance between a data point and all points in the *nearest* neighboring cluster.

Silhouette Analysis

K=2

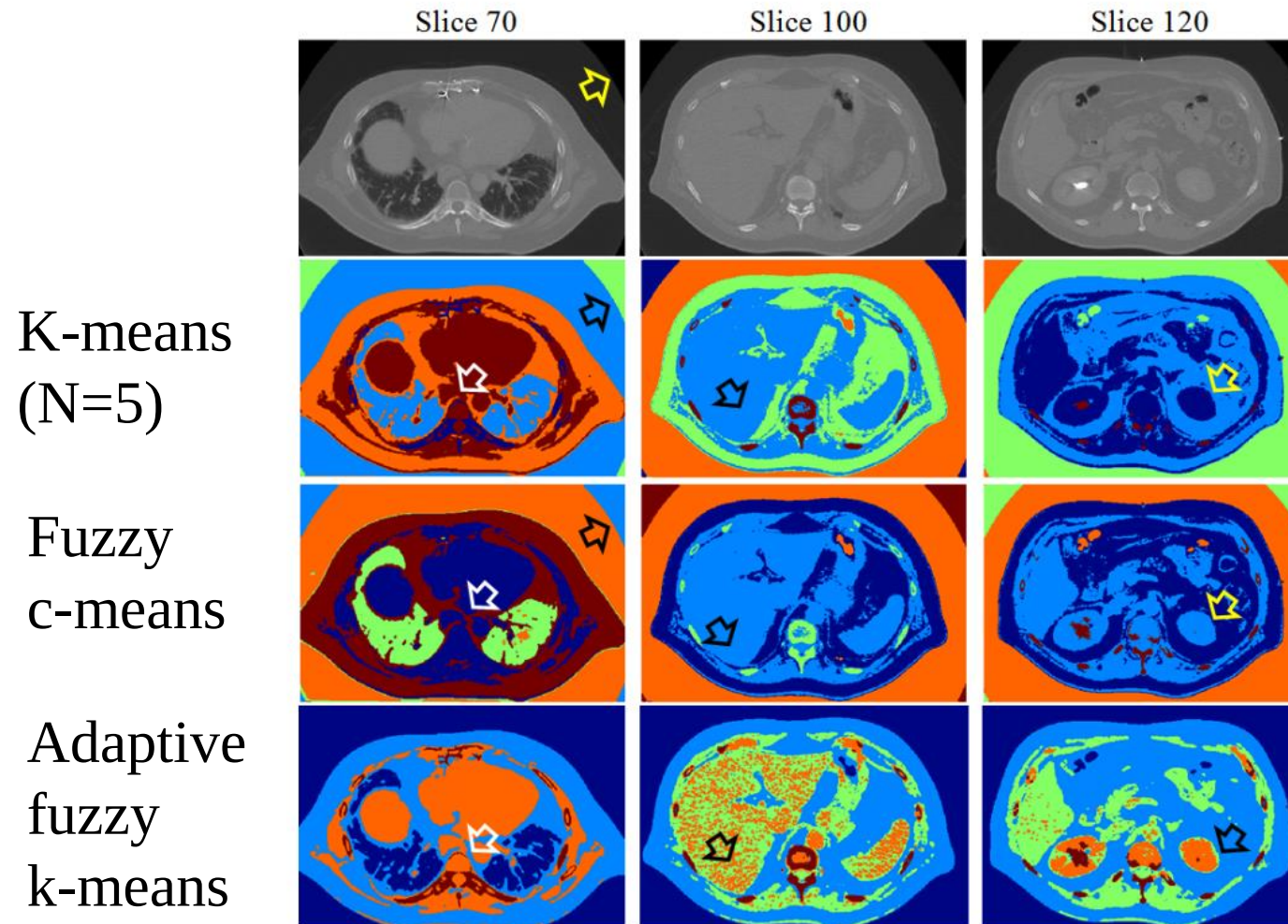
K=4



K=5

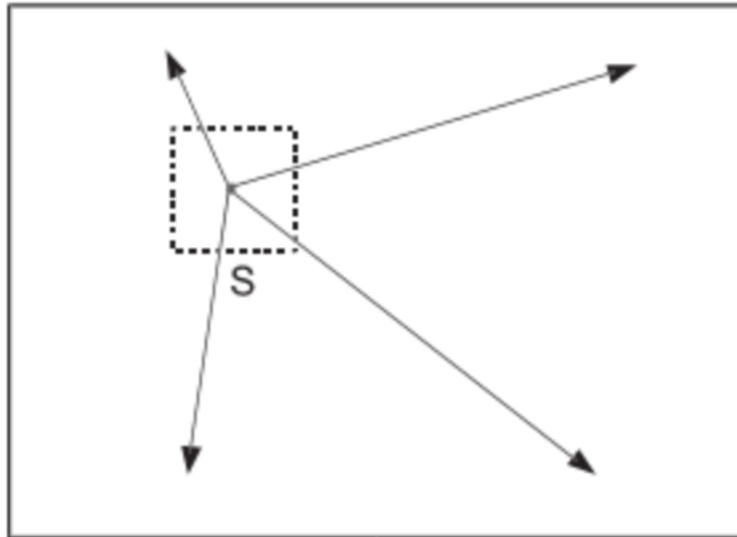
K-Means Clustering

Multi-Organ Segmentation in CT Liver-Images

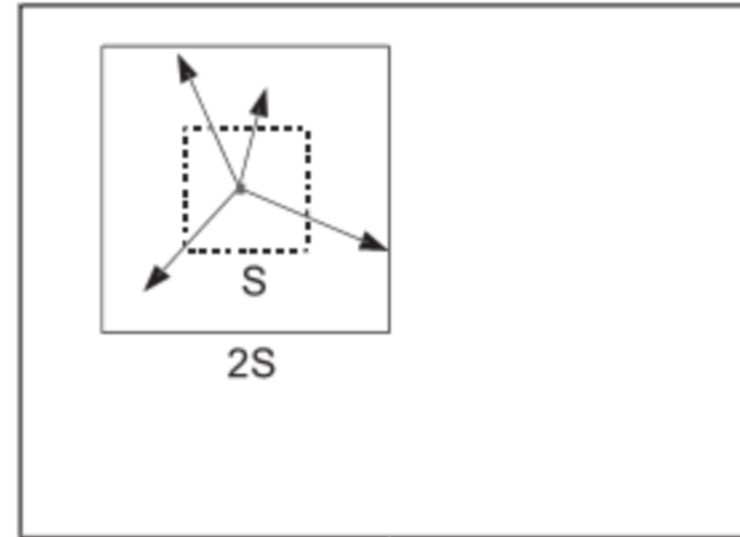


SLIC Superpixel

- Simple linear iterative clustering (SLIC)



K-mean



SLIC

SLIC Superpixel

- Initialize the algorithm

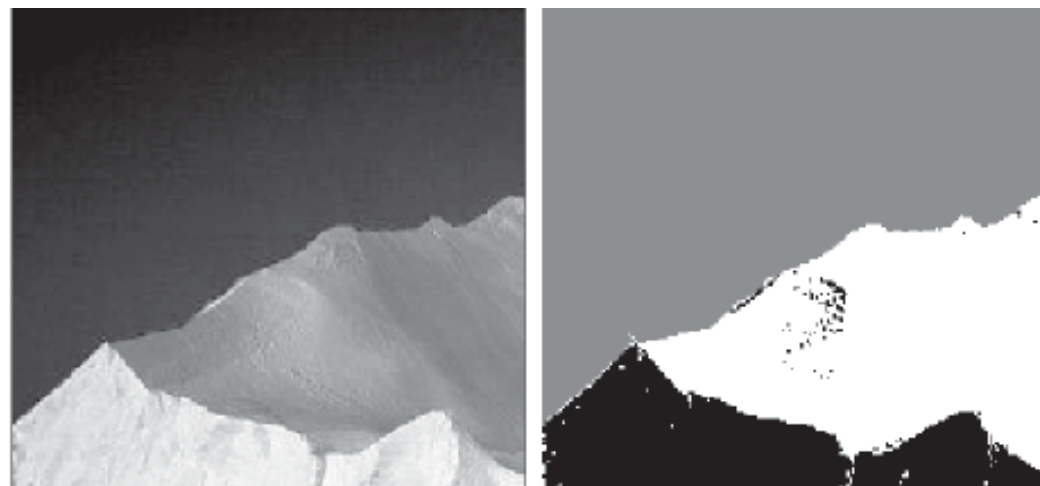
$$m_i = \{r, g, b, x, y\}$$

- Assign samples to cluster centers
- Update the cluster centers

$$m_i = \text{mean}(z), z = \{r, g, b, x, y\}_{\text{pixel}}$$

- Test for convergence
- Post-process the superpixel regions
 - Sign average value, m_i

Example



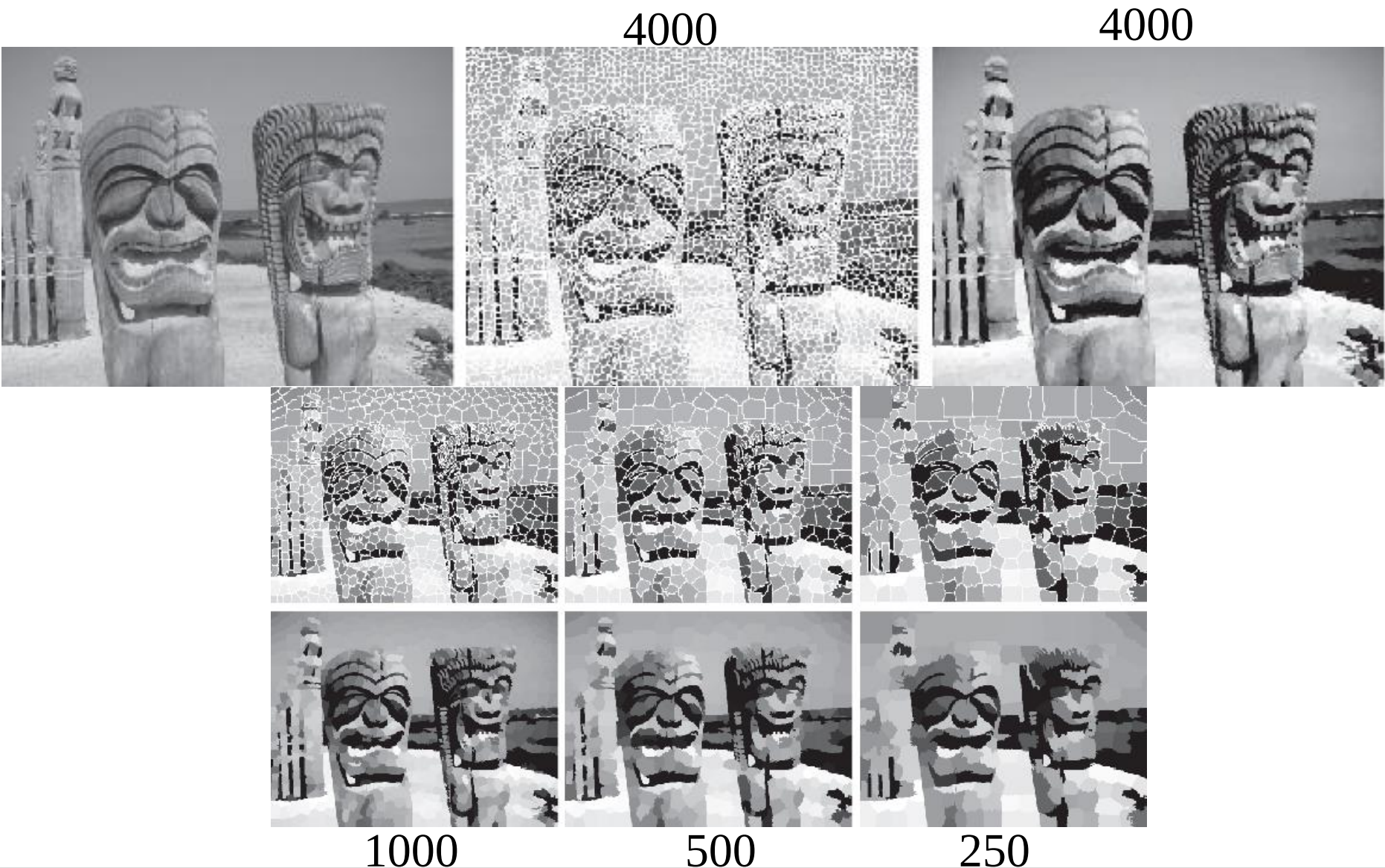
K-mean



100

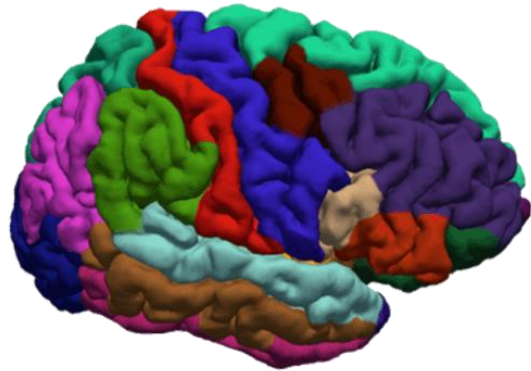
100+k-mean

Example

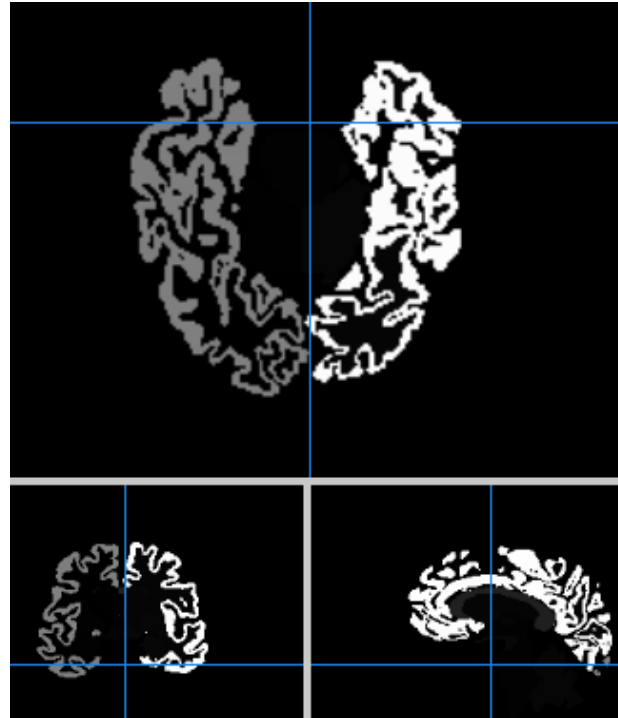


Application

- Freesurfer - Whole brain segmentation



#No.	Label Name:	R	G	B	A
0	Unknown	0	0	0	0
1	Left-Cerebral-Exterior	70	130	180	0
2	Left-Cerebral-White-Matter	245	245	245	0
3	Left-Cerebral-Cortex	205	62	78	0
4	Left-Lateral-Ventricle	120	18	134	0
5	Left-Inf-Lat-Vent	196	58	250	0
6	Left-Cerebellum-Exterior	0	148	0	0
7	Left-Cerebellum-White-Matter	220	248	164	0
8	Left-Cerebellum-Cortex	230	148	34	0
9	Left-Thalamus-unused	0	118	14	0
10	Left-Thalamus	0	118	14	0
11	Left-Caudate	122	186	220	0
12	Left-Putamen	236	13	176	0
13	Left-Pallidum	12	48	255	0
14	3rd-Ventricle	204	182	142	0
15	4th-Ventricle	42	204	164	0
16	Brain-Stem	119	159	176	0
17	Left-Hippocampus	220	216	20	0
18	Left-Amygdala	103	255	255	0
19	Left-Insula	80	196	98	0
20	Left-Operculum	60	58	210	0
21	Line-1	60	58	210	0
22	Line-2	60	58	210	0
23	Line-3	60	58	210	0

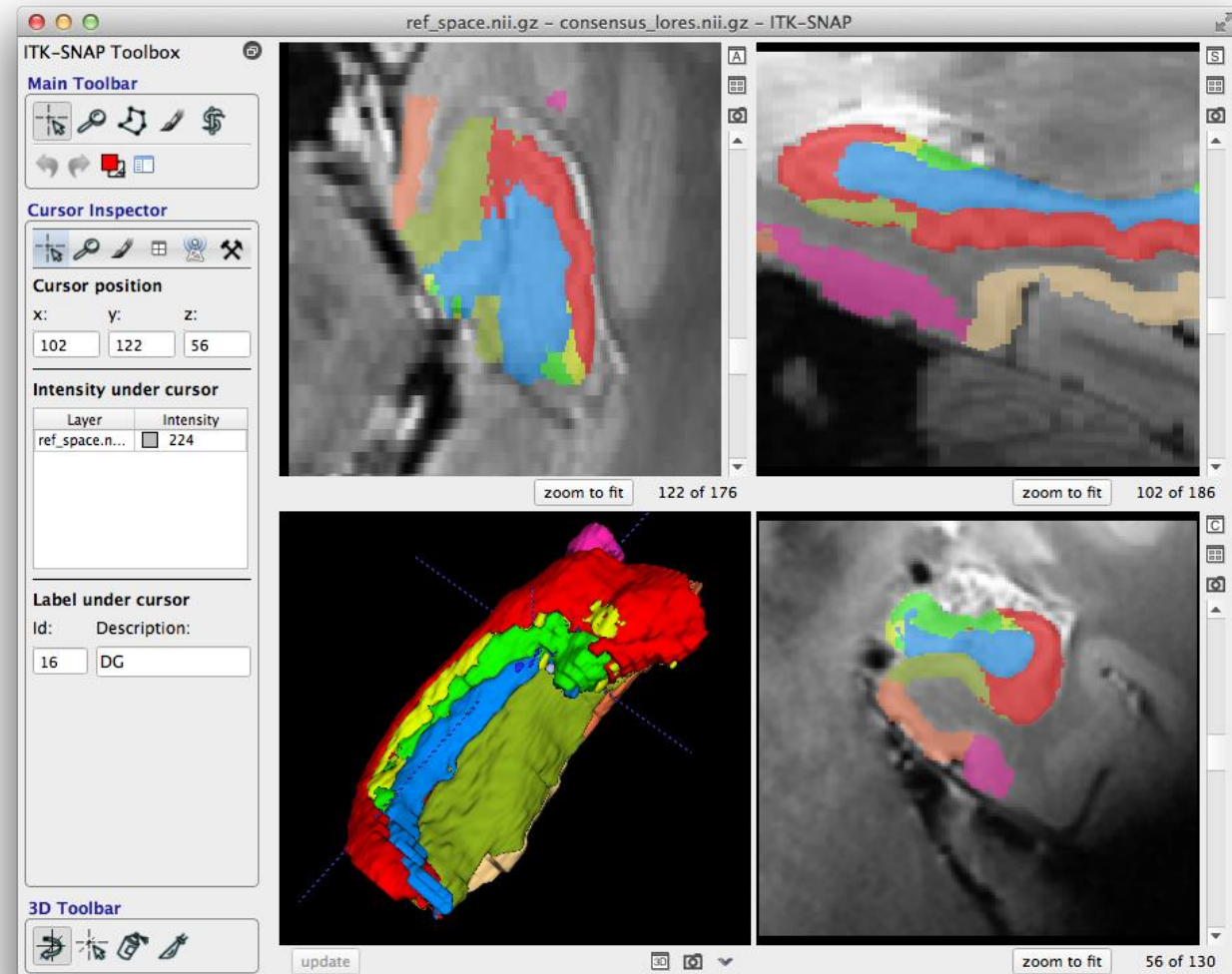


- Bayesian method
- Markov Random Field (MRF) model
- A separate model for each structure at each point in space.

Application

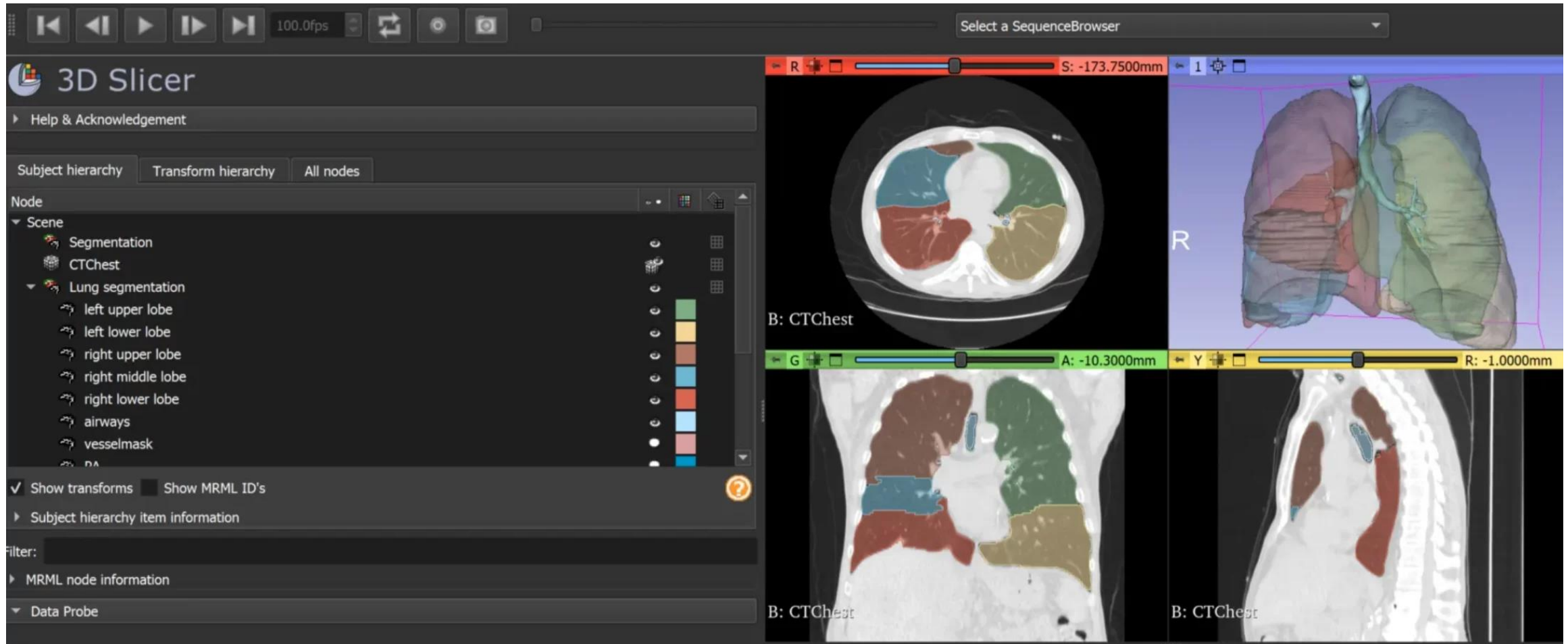


Manual+ semi-automated
segmentation



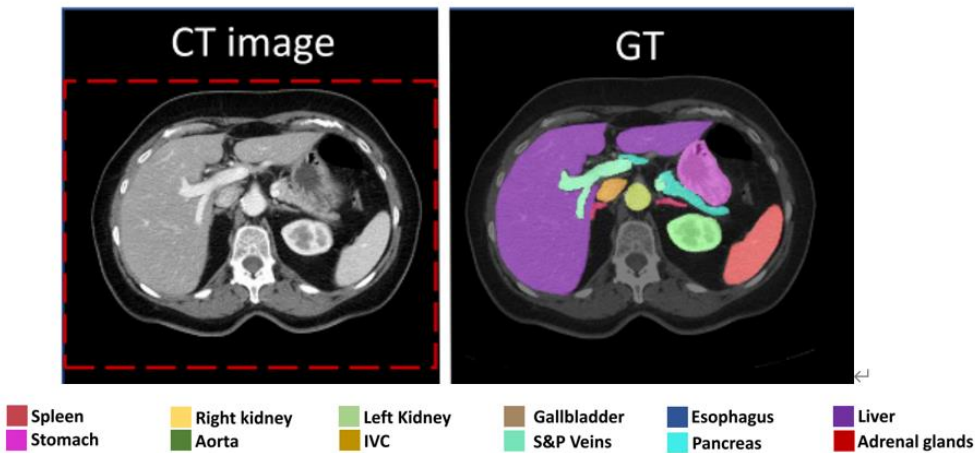
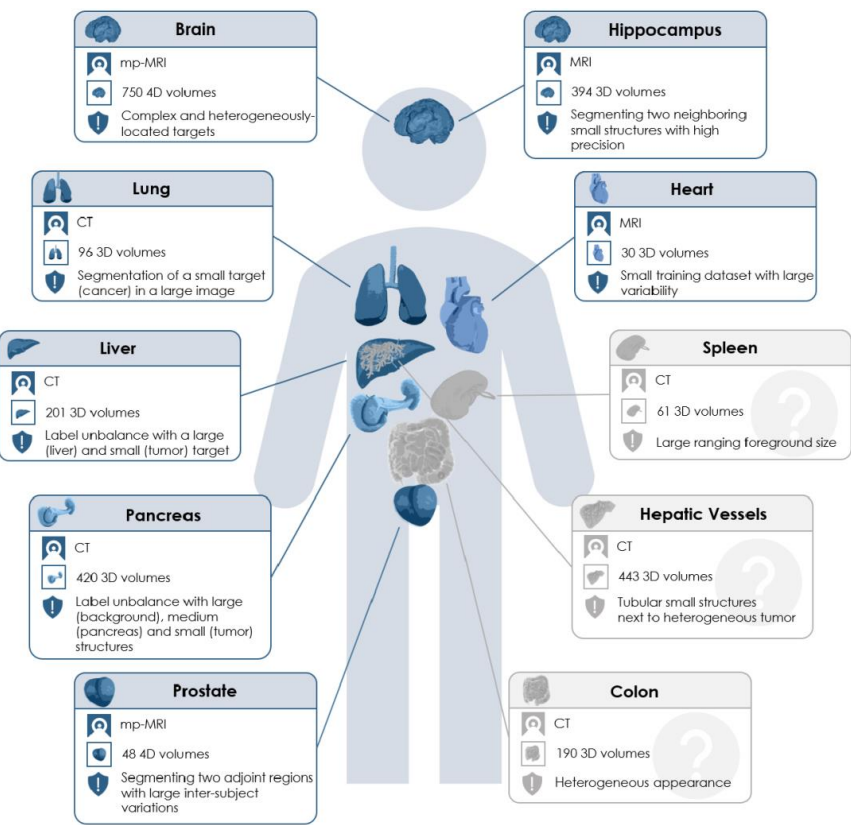
Application

Commonly used medical annotation tools for Healthcare AI (Humans in the Loop)



Application

- Dataset:
- Multi-Atlas Labeling Beyond The Cranial Vault (BTCV), CT
- Medical Segmentation Decathlon (CT, MRI)



- BraTs

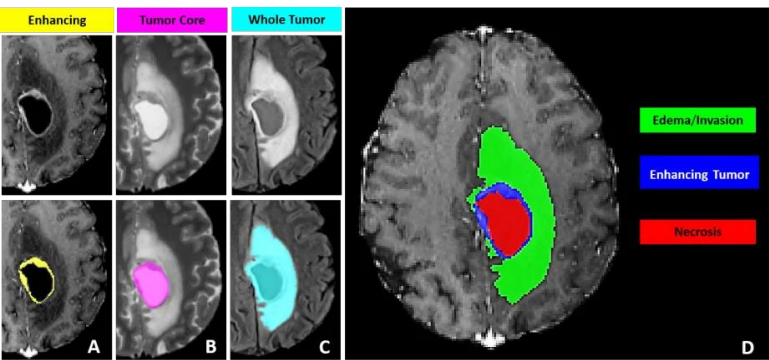


Image segmentation algorithms

- Edge– based methods
 - Point, line, edge detection filters
 - High-level edge detection algorithms
- Region-based methods
 - Thresholding
 - Region growing
- Watershed
- Clustering
 - k-mean
 - SLIC
- Application